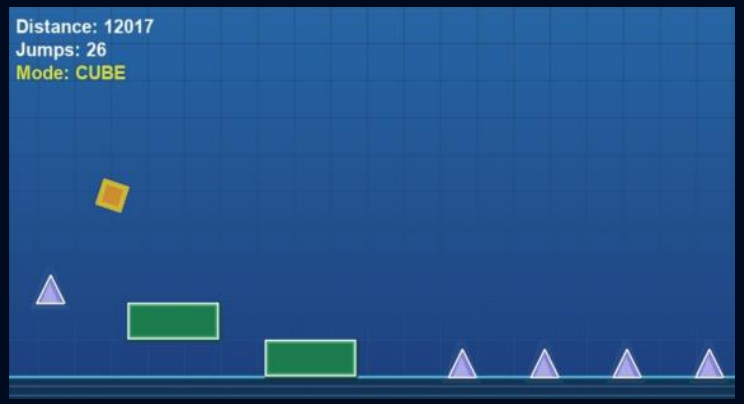
Evolutionary Algorithms

Geometry Dash

Problem Definition & Genome Representation

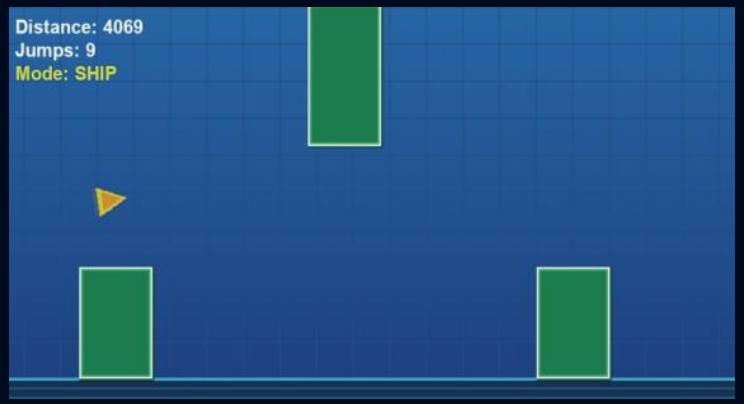
Objective:

Navigate the pre-generated level from start to finish.



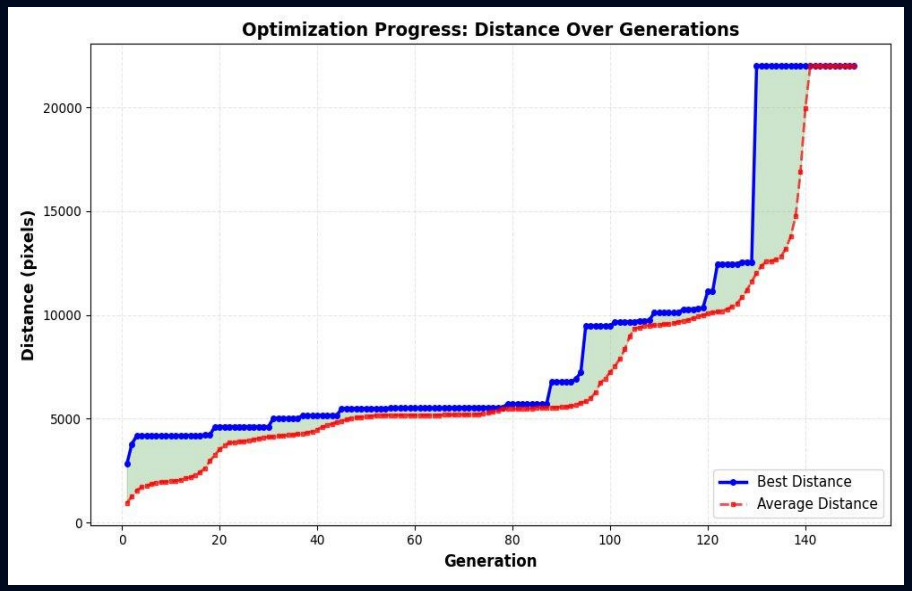
Genome:

160 float array, where even idx is time to jump and odd idx is how much jump will continue.



Transition from Cube to Ship mode.

- Cube requires impulse (single tap)
- Ship requires continuous thrust (holding).

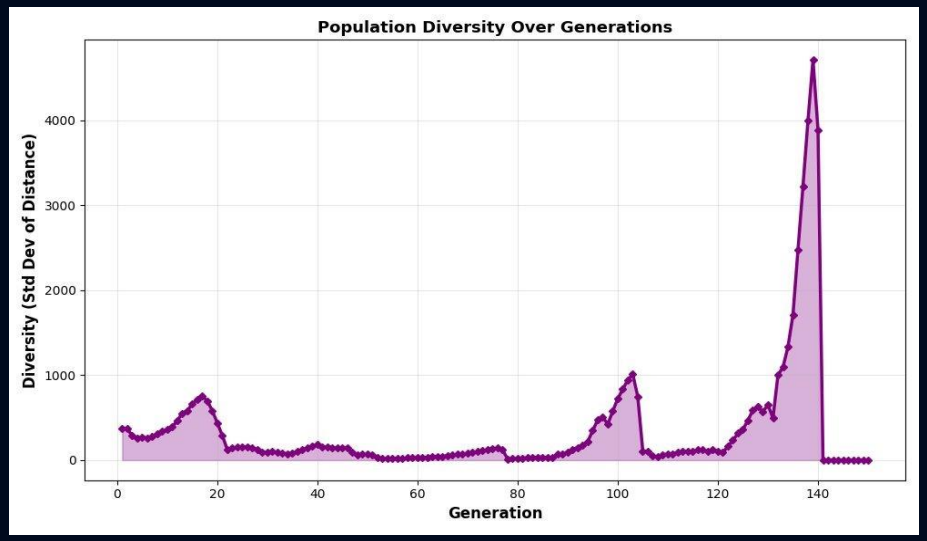


Result: Population converged at 5,500px until a random mutation discovered thrust logic.

Breakthrough: Occurred at Gen 88, leading to rapid expansion.

The Breakthrough Spike (Gen 135):

A huge increase in diversity. A small group of genomes successfully passed the Ship section and flew thousands of pixels further than the rest, creating a huge fitness gap.

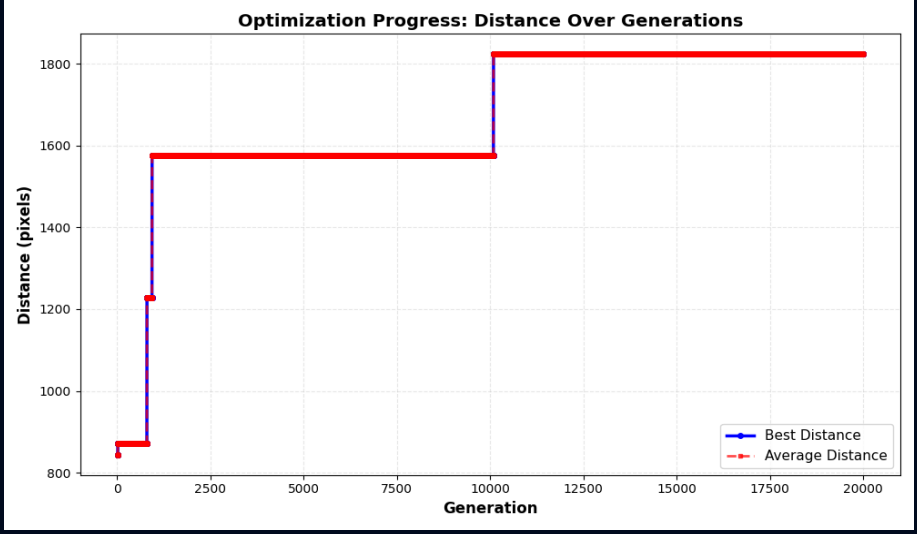


Finish (Gen 141): Diversity hits 0. The entire population has converged on the optimal solution. Every single solution is now a clone of the best performer.

Local Search, Evolutionary Algorithm & Memetic

Hill Climbing (Local Search)

- **Result:** Failed at **2,500px**.
- **Reason:** Trapped in local optima. The fitness landscape is too difficult for simple gradient-based search.



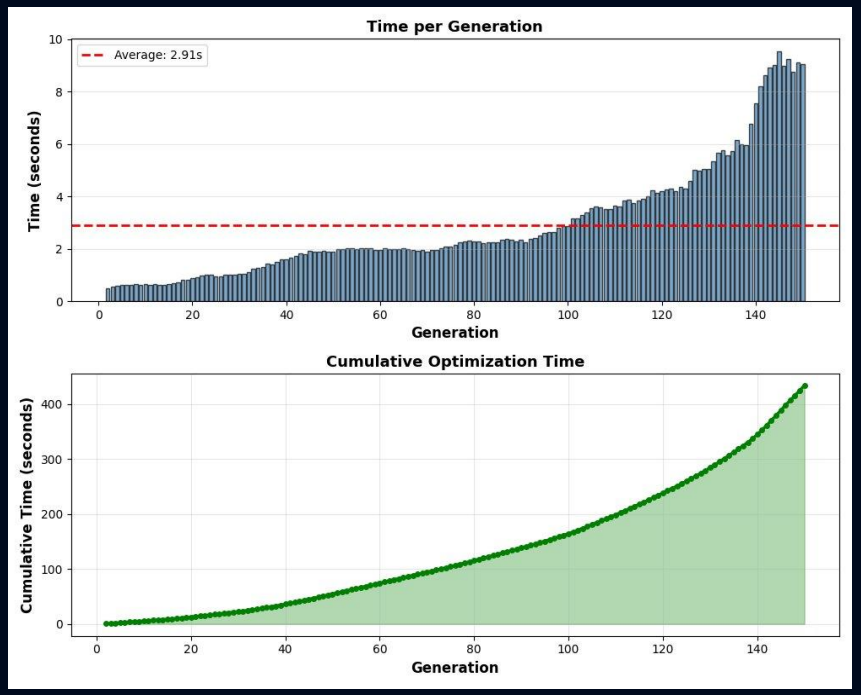
Memetic Algorithm (GA + LS)

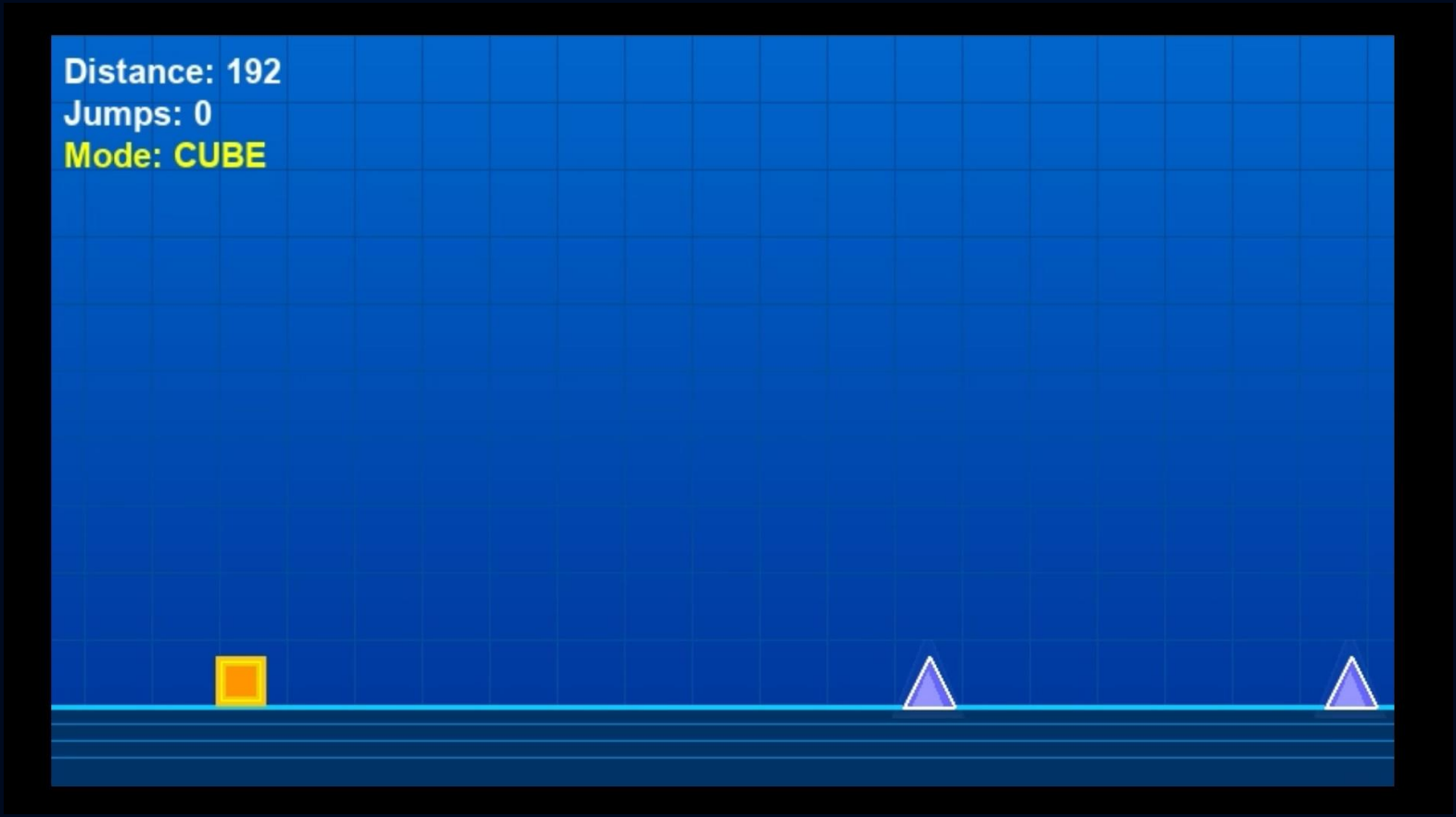
- **Result:** Reached **21,500px**.
- **Problem: Not cost-effective.** It achieved similar results to the standard EA but with longer runtime.

Total Evaluations: 67,500
Total Runtime: ~8m on PC

Final Best Individual:

- Distance: 22,009px
- Efficiency: 489px per jump
- Time: 10.35sec





Adaptability: Evolved strategies for two different physics modes.

Efficiency: The agent learned a complex strategy (jump further to minimize risk).

Thank you